

Data Model: EBS Project Management

Date: 2026-03-30
Feature: 001-ebs-project-management
Decision Ref: [research.md](#) — R1 (Separate tables per level), R2 (Sync recalculation)

Entity Relationship Overview

```
Team 1—* Member
Team *—* Project (team_projects junction)
Member 1—* Allocation
Project 1—* Phase
Phase 1—* Milestone
Milestone 1—* Epic
Milestone 1—* Allocation
Epic 1—* Story
HierarchyItem (any) 1—* ChangeLogEntry
```

Entities

Team

Field	Type	Constraints	Description
id	UUID	PK	Unique identifier
name	string(255)	NOT NULL, UNIQUE	Team display name
created_at	timestamp	NOT NULL, DEFAULT now()	Creation timestamp
updated_at	timestamp	NOT NULL, DEFAULT now()	Last update timestamp

Relationships: Has many Members. Many-to-many with Projects via [team_projects](#).

Member

Field	Type	Constraints	Description
id	UUID	PK	Unique identifier
name	string(255)	NOT NULL	Full name
email	string(255)	UNIQUE, nullable	Contact email
team_id	UUID	FK → Team.id, NOT NULL	Home team
base_allocation	decimal(5,2)	NOT NULL, DEFAULT 1.00	Base allocation (1.00 = 100%)

Field	Type	Constraints	Description
created_at	timestamp	NOT NULL, DEFAULT now()	Creation timestamp
updated_at	timestamp	NOT NULL, DEFAULT now()	Last update timestamp

Computed (not stored):

- **total_allocation**: sum of **contribution_percentage** across all Allocations where the linked milestone is Active
- **average_velocity**: mean of **velocity** across all Allocations where the linked milestone is Active
- **allocation_status**: "over-allocated" if total_allocation > 1.0, "under-utilized" if < 1.0, "balanced" if = 1.0

Relationships: Belongs to one Team. Has many Allocations.

Project

Field	Type	Constraints	Description
id	UUID	PK	Unique identifier
name	string(255)	NOT NULL	Project name
description	text	nullable	Project description
state	enum	NOT NULL, DEFAULT 'not_started'	One of: not_started, active, on_hold, completed, cancelled
total_estimated_points	integer	NOT NULL, DEFAULT 0	Aggregated from child phases
original_start_date	date	nullable	Originally planned start
original_end_date	date	nullable	Originally calculated end
original_end_date_stats	jsonb	nullable	Statistics for original calculation
actual_start_date	date	nullable	Actual start when work began
adaptive_end_date	date	nullable	Current projected end date
sort_order	integer	NOT NULL, DEFAULT 0	Display ordering
is_placeholder	boolean	NOT NULL, DEFAULT false	Auto-generated placeholder flag
created_at	timestamp	NOT NULL, DEFAULT now()	
updated_at	timestamp	NOT NULL, DEFAULT now()	

Relationships: Has many Phases. Many-to-many with Teams via [team_projects](#).

Phase

Field	Type	Constraints	Description
id	UUID	PK	
project_id	UUID	FK → Project.id, NOT NULL	Parent project
name	string(255)	NOT NULL	Phase name
description	text	nullable	
state	enum	NOT NULL, DEFAULT 'not_started'	Same 5-state enum
total_estimated_points	integer	NOT NULL, DEFAULT 0	Aggregated from child milestones
original_start_date	date	nullable	
original_end_date	date	nullable	
original_end_date_stats	jsonb	nullable	
actual_start_date	date	nullable	
adaptive_end_date	date	nullable	
sort_order	integer	NOT NULL, DEFAULT 0	Position within project
is_placeholder	boolean	NOT NULL, DEFAULT false	
created_at	timestamp	NOT NULL, DEFAULT now()	
updated_at	timestamp	NOT NULL, DEFAULT now()	

Relationships: Belongs to Project. Has many Milestones.

Milestone

Field	Type	Constraints	Description
id	UUID	PK	
phase_id	UUID	FK → Phase.id, NOT NULL	Parent phase
name	string(255)	NOT NULL	
description	text	nullable	
state	enum	NOT NULL, DEFAULT 'not_started'	Same 5-state enum

Field	Type	Constraints	Description
total_estimated_points	integer	NOT NULL, DEFAULT 0	Aggregated from child epics
efficiency_factor	decimal(3,2)	NOT NULL, DEFAULT 1.00	Buffer (0.10–1.00)
is_parallel	boolean	NOT NULL, DEFAULT false	Can overlap adjacent milestones
original_start_date	date	nullable	
original_end_date	date	nullable	
original_end_date_stats	jsonb	nullable	Calculation stats for original estimate
actual_start_date	date	nullable	
adaptive_end_date	date	nullable	
sort_order	integer	NOT NULL, DEFAULT 0	Position within phase
is_placeholder	boolean	NOT NULL, DEFAULT false	
created_at	timestamp	NOT NULL, DEFAULT now()	
updated_at	timestamp	NOT NULL, DEFAULT now()	

Relationships: Belongs to Phase. Has many Epics. Has many Allocations.

Epic

Field	Type	Constraints	Description
id	UUID	PK	
milestone_id	UUID	FK → Milestone.id, NOT NULL	Parent milestone
name	string(255)	NOT NULL	
description	text	nullable	
state	enum	NOT NULL, DEFAULT 'not_started'	Same 5-state enum
total_estimated_points	integer	NOT NULL, DEFAULT 0	Aggregated from child stories
original_start_date	date	nullable	
original_end_date	date	nullable	
actual_start_date	date	nullable	
adaptive_end_date	date	nullable	

Field	Type	Constraints	Description
sort_order	integer	NOT NULL, DEFAULT 0	Position within milestone
is_placeholder	boolean	NOT NULL, DEFAULT false	
created_at	timestamp	NOT NULL, DEFAULT now()	
updated_at	timestamp	NOT NULL, DEFAULT now()	

Relationships: Belongs to Milestone. Has many Stories.

Story

Field	Type	Constraints	Description
id	UUID	PK	
epic_id	UUID	FK → Epic.id, NOT NULL	Parent epic
name	string(255)	NOT NULL	
description	text	nullable	
state	enum	NOT NULL, DEFAULT 'not_started'	Same 5-state enum
estimated_points	integer	NOT NULL, DEFAULT 0	Leaf-level points (not aggregated)
original_start_date	date	nullable	
original_end_date	date	nullable	
actual_start_date	date	nullable	
adaptive_end_date	date	nullable	
sort_order	integer	NOT NULL, DEFAULT 0	Position within epic
is_placeholder	boolean	NOT NULL, DEFAULT false	
created_at	timestamp	NOT NULL, DEFAULT now()	
updated_at	timestamp	NOT NULL, DEFAULT now()	

Relationships: Belongs to Epic. Leaf node — no children allowed.

Allocation

Field	Type	Constraints	Description
id	UUID	PK	

Field	Type	Constraints	Description
member_id	UUID	FK → Member.id, NOT NULL	Allocated resource
milestone_id	UUID	FK → Milestone.id, NOT NULL	Target milestone
velocity	decimal(8,2)	NOT NULL	Story points per week if 100% dedicated
contribution_percentage	decimal(5,2)	NOT NULL	Actual % contribution (e.g., 0.50 = 50%)
fto	decimal(8,2)	nullable	Finish-Time Offset (user-provided estimate)
created_at	timestamp	NOT NULL, DEFAULT now()	
updated_at	timestamp	NOT NULL, DEFAULT now()	

Unique constraint: (member_id, milestone_id) — a member can only have one allocation per milestone.

Computed (not stored):

- **effective_velocity:** velocity × contribution_percentage

Relationships: Belongs to Member. Belongs to Milestone.

ChangeLogEntry

Field	Type	Constraints	Description
id	UUID	PK	
item_type	string(50)	NOT NULL	'project', 'phase', 'milestone', 'epic', 'story'
item_id	UUID	NOT NULL	ID of the affected hierarchy item
previous_end_date	date	nullable	End date before the change
new_end_date	date	nullable	End date after the change
reason	text	NOT NULL	Human-readable change description
created_at	timestamp	NOT NULL, DEFAULT now()	When the change occurred

Index: (item_type, item_id, created_at DESC) for efficient change log retrieval.

team_projects (Junction Table)

Field	Type	Constraints	Description
team_id	UUID	FK → Team.id, NOT NULL	
project_id	UUID	FK → Project.id, NOT NULL	

Primary Key: (team_id, project_id)

State Enum

All hierarchy items use the same state values:

Value	DB Value	Calculative	Description
Not Started	not_started	No	Item created but work not begun
Active	active	Yes	Work in progress — included in velocity/allocation calculations
On Hold	on_hold	No	Paused — excluded from calculations
Completed	completed	No	Finished — excluded from calculations
Cancelled	cancelled	No	Abandoned — excluded from calculations

Validation Rules

1. **Hierarchy enforcement:** A story cannot have children. Epic → Story, Milestone → Epic, Phase → Milestone, Project → Phase. No cross-level parenting.
2. **Allocation uniqueness:** A member can have at most one allocation per milestone.
3. **Efficiency factor range:** Must be between 0.10 and 1.00 inclusive.
4. **Contribution percentage:** Must be between 0.01 and 1.00 inclusive.
5. **Sort order:** Non-negative integer. System manages reordering automatically.
6. **Placeholder replacement:** When a real item is created under a parent that has a placeholder, the placeholder with the highest sort_order is replaced first.

Aggregation Rules

1. **Points roll-up:** Epic.total_points = sum(Story.estimated_points). Milestone.total_points = sum(Epic.total_points). Phase → Project similarly.
2. **End-date calculation** (Milestone level):
 - $\text{effective_velocity} = \text{sum}(\text{allocation.velocity} \times \text{allocation.contribution_percentage}) \times \text{milestone.efficiency_factor}$
 - $\text{duration_weeks} = \text{milestone.total_estimated_points} \div \text{effective_velocity}$ (if effective_velocity > 0, else indeterminate)
 - $\text{adaptive_end_date} = \text{milestone.start_date} + \text{duration_weeks}$

3. **Date inheritance:** Child start_date defaults to parent start_date unless explicitly set.
4. **Sequential milestones:** Within a phase, milestone[n].start_date = milestone[n-1].adaptive_end_date (unless marked parallel).
5. **Gap-filling:** If milestone[n].adaptive_end_date < milestone[n+1].original_start_date and both are sequential, insert placeholder milestone to fill the gap.